

Exam 1 Review (Jeopardy Game Summary)

Data Mining General: fundamental data mining concepts, including different sources and types of data, data understanding, exploration, visualization, data preprocessing and cleaning techniques (including but not limited to missing values, outliers, normalization), data mining tasks, data mining steps.

Classification General: the foundational principles of classification in machine learning as a supervised predictive learning process. Topics included binary and multi-class classification, classification evaluation measures and metrics (including confusion matrix, accuracy, recall, precision, f1-score), and common classification algorithms.

Decision Tree: decision tree algorithms, how they're constructed, the criteria for node splitting (e.g., impurity or entropy), interpreting decision tree, splitting and pruning in decision trees, max-depth and its effect on training/fitting/overfitting.

Naive Bayes: Naive Bayes classification method, the core assumptions (feature independence), the Bayes' theorem, Laplace smoothing, and practical applications, such as spam detection, and how to use the Bayes' theorem to calculate conditional probability.

K-Nearest Neighbor: the K-Nearest Neighbor (K-NN) algorithm in-depth, distance/similarity metrics (e.g., Euclidean distance), the choice of K and its impact on the model performance (test and train), and K-NN disadvantages such as being computationally expensive and the curse of dimensionality, among others.

Generalization and Overfitting: concepts of generalization (model performance on unseen data) and overfitting (model fitting to noise), techniques to mitigate overfitting by watching the model's performance on both train and test sets.

Lab and Python Code (Pandas and Scikit-learn): coding tasks using Python, Pandas, and Scikit-learn. Topics included data preprocessing, feature selection, and building machine learning models for data analysis and prediction. You should expect more detailed questions (similar to our labs) in the exam.

This is just a summary of the topics, please go through all six of the lectures and six labs that we've completed up to this point.